



ATP Tennis Match Predictor

Select two players and match context to get predicted win probabilities from a Random Forest model trained on ATP data (2000–2024).

Player 1

Select Player 1

Zverev A. ▾

ATP Ranking (Player 1)

1 - +

Player 2

Select Player 2

Shelton B. ▾

ATP Ranking (Player 2)

8 - +

Match Context

Surface

Hard ▾

Round

The Final ▾

Series / Tournament tier

Grand Slam ▾

Court

Outdoor ▾

Best of

☐ 3 ☒ 5

Predict

Prediction

Zverev A.

37.1%

Shelton B.

62.9%



Predicted winner: **Shelton B.**

▼ Match details used

```
{
  "Player 1" : "Zverev A."
  "Rank 1" : 1
  "Player 2" : "Shelton B."
  "Rank 2" : 8
  "Surface" : "Hard"
  "Round" : "The Final"
  "Series" : "Grand Slam"
  "Court" : "Outdoor"
  "Best of" : 5
}
```

▼ ⓘ How it works

Model: Random Forest (200 trees, trained on 2000–2024 ATP match data)

Features used:

- ATP rankings of both players (`Rank_1` , `Rank_2`)
- Engineered ranking features: `rank_diff` , `rank_ratio` , `log_rank_diff`
- Match context: Surface, Round, Series/Tournament tier, Court type, Best of

What's intentionally excluded:

- Player names (never enter the model — only used to look up rankings)
- Betting odds (post-market information — data leakage)
- Match scores (post-match information — data leakage)

Validation (2025): ~61% accuracy, ROC-AUC ~0.65 **Holdout (2026):** ~61% accuracy, ROC-AUC ~0.65

ATP ranking is by far the strongest predictor, accounting for ~85% of the model's feature importance. Match context (surface, round, series) adds a meaningful but smaller signal.